

EXERCISES

1. Prove by induction that:
 - a) For every integer $n \geq 2$, $n^3 - n$ is always divisible by 3.
 - b) For every natural number r : $5^n - 1$ is always divisible by 4.
2. Prove by induction that $\sqrt{3}$ is irrational.
3. Show that $f: R \rightarrow R, f(x) = \frac{5x+3}{8}$ is a bijection and find its inverse.
4. Prove that for any integer x , $x^2 + x + 1$ is odd.
5. The final exam of discrete structures for computing consists of two sections: Section A and section B. Section B consists of 3 questions (Q2, Q3 and 4) where students have to choose any two questions from their choice. 30 student attempted Q2, 25 students attempted Q3 and 19 students attempted Q4. Find the total number of students if 18 students choose Q22 and Q3, 12 choose Q2 and Q4 and 7 choose Q3 and Q4.
6. Using direct proof, prove that the square of any odd number is an odd number.
7. A standard license plate motorbike consists of two letters followed by three numbers and ended with one letter (All plates containing 3 zeros are not included). All plates must be started with R. Letter O is not used in order to avoid confusion with number zero. How many different standard license plates are possible?
8. What is the meaning of the following FOL formulas?

a) $\exists x, Bought(Kelly, x)$

b) $\forall x (eat(mary, x) \rightarrow eat(tite, x))$

9. Find the coefficient of $x^{14}y^{13}$ in $(x - y)^{27}$

10. Using two different ways, show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (q \rightarrow r)$ is tautology.

11. In computer programming, the name of variable is a string of one or two or three alphanumeric characters where uppercase and lower case are distinguished. A variable name cannot be a reserved word (keyword) and must begin with a letter. All keywords are lower letters; computer programming consist of the following keywords: if, else, return, switch, case, default, for, do, while, break, continue, int, float, char, void, struct, typed, union, enum, sizeof, and extern. How many different variable names are there in computer programming?

12. Use truth table to show that $A \rightarrow B$ and $\neg A \vee B$ are logically equivalent

13. I) Given that $f(x) = 3x + 7$ and $g(x) = 2x^2 + 3$ find:

a) $(f \circ g)(x)$

b) $(g \circ f)(x)$

II) Compute $f \circ g(x)$ if $g(x) = 2x^2 + 2$ and $f(x) = 2x^6 + 3$

14. Using induction proof, show that $1+2+4+8+ \dots + 2^n = 2^{n+1} - 1$

15. A bank has 214 customers. 189 of them have checking account, 73 have normal saving accounts and 114 have money market saving accounts. 69 customers have checking accounts and normal saving accounts. No customer is allowed to have a normal account and a money market savings account.
- a) How many customers have checking accounts and money market savings accounts?
 - b) How many customers have a checking account, but no saving account at all?
16. Construct a truth table of $P \rightarrow \neg P \wedge P \vee \neg P$
17. Without using truth table, prove that $(\neg Q \wedge P) \rightarrow (Q \rightarrow P)$ is tautology
18. Show that the premises “It is sunny this afternoon and it is colder than yesterday,” “We will go swimming only if it is sunny,” “If we do not go swimming, then we will take a canoe trip,” and “If we take a canoe trip, then we will be home by sunset” lead to the conclusion “We will be home by sunset.”
19. A total of 1232 students have a course in Kinyarwanda, 879 have taken a course in French, and 114 have taken a course in English. Further, 103 have taken course in both Kinyarwanda and French, 23 have taken course in both Kinyarwanda and English and 14 have taken course in both French and English. If 2092 students have taken at least one of Kinyarwanda, French and English.

How many students have taken a course in all three languages?

20. Prove that $m + n$ and $n + p$ are even integers where m , n and p are integers, then $m + p$ is even. What kind of proof did you use?

21. Let P , Q and R be the propositions.

P : You get A on the final exam

Q : You do every exercise in this book

R : You get an A in this class

Write the following using P , Q and R and logical

connectives. **You get an A in this class, but you do not do every exercise in this book**

22. Alice, Bob, Kamila, Eric and Dan are making plans for spring break. They go to the travel agency, but they are only two tickets left. Alice will only go if Bob goes too. Dan will only go if Kamila goes too. Bob has found out that he has to work on this issue, so he cannot go referring to text above, write the proposition logic formulas corresponding to statements

Below:

a) Alice will go if Bob goes

b) Dan goes only if Kamila goes

c) Alice and Dan will go if and only if Bob do not goes

d) Eric will go if Alice, Bob, Kamila and Dan decide not to go

23. Suppose that the universe of discourse of the atomic formula $P(x, y)$ is $(1, 2, 3)$. Write out the following propositions using disjunction and conjunction

- a) $\exists xP(x, 2)$
- b) $\forall xP(3, y)$
- c) $\forall x\forall yP(x, y)$
- d) $\exists x\exists yP(x, y)$

24. In arithmetic progression, the sum of the first 10 terms is 400, the sum of the next ten terms is 1000. Find the common difference and the first term.
25. Construct the truth table for each of the proposition
- a) $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$
 - b) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
26. In computer programming the name of a variable is a string of one or two alphanumeric characters where uppercase and lowercase are not distinguished. A variable name cannot be reserved word(keyword) and must begin with a letter. How many different variable names are there if there are six keywords in computer programming?
27. A computer company receives 350 applicants from computer graduates for a job planning a line of new web server suppose that 220 applicants of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business.
28. Let f and g be the function from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$
- a) Find $f - g(x)$

- b) What is the composition of f and $g(f \circ g(x))$?
29. Let A and B be sets. A represents all prime numbers between 20 and 40 and B represents all divisors of 92.
- List all terms in each set.
 - Calculate $A \cap B$, $|A \cup B|$, $P(A)$
30. Using basic set identities, prove that $A \cap B = B - (B - A)$
31. Using direct proof, prove that the product of two numbers is an odd number
32. Using binomial coefficients, calculate the expansion of $(x + y)^5$
33. An arithmetic progression has 3 as its first term. Also, the sum of the first 8 terms is twice the sum of the first 5 terms. Find the common difference.
34. Let $A = \{x | x \in \mathbb{Z}^+ \text{ and } x \text{ is a prime number less than } 40\}$ and $B = \{x | x \text{ is a divisor of } 70\}$.
- Define the elements of A and B
 - Calculate $A \cap B$, $|A \cup B|$ and $|P(B)|$
35. Using two techniques, prove that $A - (B \cup C)$ and $(A - B) \cap (A - C)$ are equivalent.
36. How many permutations of the letters MUSANZE contain the string MUS?
37. In a school, 30 teachers teach C programming or C++ programming. 21 teach C programming and 6 teach C programming and C++ programming. Find the number of teachers who can teach C++ programming.

38. Find the bitwise OR, bitwise AND, and bitwise XOR of each of these pairs of bit strings. 11110000, 10101010
39. The Fibonacci sequence, $f_0, f_1, f_2, \dots$, is defined by the initial conditions $f_0=0, f_1=1$, and the recurrence relation $f_n = f_{n-1} + f_{n-2}$ for $n=2,3,4,\dots$. Find the f_6
40. In a group of 100 students, 73 like programming and 45 like networking. Find the number of students who like programming only and the students who like networking only.
41. How many rows and columns appear in a truth table for each of these compound propositions?
 a) $Q \vee P \vee \neg S \vee \neg R \vee \neg Q \vee U$
 b) $(P \wedge R \wedge T) \leftrightarrow (Q \wedge T)$
42. Find $f(5)$ if f is defined recursively by $f(0) = f(1)$ and $f(n+1) = f(n) - f(n-1)$ for $n > 1$.
43. Using induction proof, show that if n is a positive integer, then $1 + 2 + \dots + n = \frac{n(n+1)}{2}$
44. Suppose that on an island there are three types of people, knights, knaves, and normal (also known as spies). Knights always tell the truth, knaves always lie, and normal sometimes lie and sometimes tell the truth. Detective questioned three inhabitants of the Island-Amy, Brenda, and Claire – as part of the investigation of a crime. The detective knew that one of the three committed the crime, but not which one. They also knew that the was knight
45. Determine the truth value of each of these statements if the domain consists of all integers

- a) $\exists n(2n = 3n)$
- b) $\forall n(n + 1 > n)$
- c) $\forall n(n^2 \geq n)$
- d) $\exists n(n = -n)$

46. A soccer club has 8 female and 7 male members. For today's match, the coach wants to have 6 females and 5 male players on the grass. How many possible configurations are there?
47. Which suppose that smartphone A has 256 MB RAM and 32 GB ROM, and the resolution of its camera is 8MP, smartphone B has 288 MB RAM and 64 GB ROM, and the resolution of its camera is 4 MP, and smartphones C has 128 MB RAM and 32 GB ROM, and the resolution of its camera is 5MP. Determine the truth value of the three smartphone.
- a) Smartphone B has the most RAM of these three smartphones.
 - b) Smartphone C has more ROM or higher resolution.
 - c) Smartphone B has more RAM, more ROM and higher resolution camera than smartphone A.
48. In the version of the computer language BASIC, the name of a variable is as string of one, two or three alphanumeric characters, where uppercase and lowercase letters are not distinguished. (An alphanumeric character is either one of the 26 English letters or one of the 10 digits). Moreover, a variable name must begin with a letter and

must be different from the 10 digits of two or three characters that are reserved for programming use. How many different variable names are there in this version of BASIC?

49. Construct truth table for $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$
50. Using example, show that composition of function is not associative.
51. List the members of these sets.
 - a) $A = \{x | x \in R \text{ and } x^2 + x - 12 = 0\}$ where R set represent all real numbers
 - b) $B = \{x | x \text{ is a positive integer less than } 12\}$
 - c) $C = \{x | x \text{ is the square of an integer and } x < 100\}$
 - d) Find $A \cap B \cap C$, $(A \cap B) \cup C$, $(A \cap B) \cup (B \cap C)$
52. What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the value of this statement if the domain is consist of all integers?
53. How many students, each of whom comes from one of the 30 districts, must be enrolled at the university to guarantee that there are at least 100 who come from the same district?
54. Given $A = \{e, f, g\}$ and $B = \{2, 1\}$, what is their Cartesian product $A \times B$?
55. Which of these sentences are proposition? What are truth values of those that are proposition?
 - a) Kigali is the capital of Huye
 - b) Kampala is the capital of Tanzania
 - c) $X + 2 = 11$

- d) Answer this question
 - e) $X + y + z = 0$
 - f) Where are going
 - g) $5 - 0 = 9$
 - h) Do the work quickly
 - i) $X > 0, x = \{All\ positive\ integers\}$
56. Construct a truth table for $((p \rightarrow q) \rightarrow r) \rightarrow s$.
57. Show that this conditional statement is a tautology by using truth tables.

$$[\neg p \wedge (p \vee q)] \rightarrow q$$
58. Let P and Q be the propositions
 P: Your car is out of gas
 Q: You can't drive your car.
 Write the following propositions using P and Q and logical connectives.
- a) Your car is not out of gas
 - b) You can't drive your car if it is out of gas
 - c) Your car is not out of gas if you can drive it
 - d) If you can't drive your car, then it is out of gas.
59. Determine whether each of the following implications is true or false.
- a) If 0.9 is an integer, then $1+0.5=3$.
 - b) If birds can fly, then $1+1=3$.
 - c) If $5 \geq 2$ then cows can fly.
 - d) If $3*5=15$ then $1+2 \leq 3$.
60. Construct a truth table for each of the following implications.

- a) $P \wedge \neg P$
- b) $(P \vee Q) \rightarrow Q$
- c) $(PQ) \leftrightarrow (\neg Q \rightarrow \neg P)$

61. Express the following arguments using the symbols indicated. What rules of inference are used in each of them?

- a) If the teens like it, then the sales volume will go up;
Either the teens like it or the store will close; The sales volume will not go up. Therefore, the store will close.
Symbols to be used: The teens like it (T). The sales volume will go up (S). The store will close (C)
- b) It is not the case that if there is not a lot of sun, then there is enough water, nor is it true that either there is a lot of rain or the crop is good. Therefore, there is not enough water and the crop is not good. Symbols to be used: There is not a lot of sun (S). There is enough water (W). There is a lot of rain (R). The crop is good (C).
- c) If flowers are colored, they are always scented; I don't like flowers that are not grown in the open air; All flowers grown in the open air are colored. Therefore, I don't like any flowers that are scentless. Symbols to be used: Flowers are colored (C). Flowers are scented (S). I like flowers (L). Flowers are grown in open air (O).
- d) No animals, except giraffes, are 15 feet or higher: There are no animals in this zoo that belong to anyone but me; I have no animals less than 15 feet high. Therefore, all animals in this zoo are giraffes. Symbols to be used:

Animals are giraffes (G). Animals are 15 feet or higher (F). Animals are in this zoo (Z). Animals belong to me (M)

62. Given the following sets.

$A = \{x | x \in N \text{ and } 130 \bmod x = 0 \text{ and } x \text{ is less than or equal } 130\}$ where N set represents integers

$B = \{x | x \text{ is a prime number less than } 30\}$

a) List members of these sets

b) Find $A \cap B$, $|P(B)|$ and $A - B$

63. In C programming, upper letters and lower letters are different. The name of a variable is a string of one or two alphanumeric characters. A variable name cannot be a reserved word (keyword) and must begin with a letter. How many different variable names are there if there are 12 keywords in computer programming?

64. Write a recursion function of Greatest Common Divisor between two numbers and Calculate GCD (240,360)

65. Find the coefficient of $a^{25}b^{22}$ in $(a + b)^{45}$

Question 1(3+3+3+3+2+3+3)

1.1 Find MST and adjacency matrix of the following graph



